

Lecture 4

Recall: Let $Mdx + Ndy = 0$. If it is not exact and

$\frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right)$ is a function of x then

$$\text{IF} = e^{\int \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) dx} \quad \text{and}$$

if $\frac{1}{M} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right)$ is a function of y then

$$\text{IF} = e^{\int \frac{1}{M} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) dy}.$$

Ex. 1 $(8xy - 9y^2) + (2x^2 - 6xy)y' = 0.$

Soln: Here $M = 8xy - 9y^2$ and $N = 2x^2 - 6xy$

$$\therefore \frac{\partial M}{\partial y} = 8x - 18y, \quad \frac{\partial N}{\partial x} = 4x - 6y$$

Clearly, $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, thus the given eq is not exact.

$$\therefore \frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = 8x - 18y - 4x + 6y = 4x - 12y = 4(x - 3y)$$

$$\Rightarrow \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = \frac{1}{2x(x-3y)} \cdot 4(x-3y) = \frac{2}{x} \quad (\text{function of } x).$$

$\therefore$ There exist an integrating factor given by

$$\text{IF } F = e^{\int \frac{2}{x} dx} = e^{2 \ln x} = e^{\ln x^2} = x^2.$$

Multiply the given eq by the integrating factor x^2 .

$$x^2(8xy - 9y^2) + x^2(2x^2 - 6xy)y' = 0.$$

This eq is exact. Thus $\exists$ a function u s.t

$$\frac{\partial u}{\partial x} = x^2(8xy - 9y^2) \quad \& \quad \frac{\partial u}{\partial y} = x^2(2x^2 - 6xy).$$

$\Rightarrow$ Partially integrate the first eq w.r to x ,

$$u = 8y \frac{x^4}{4} - 9y^2 \frac{x^3}{3} + \phi(y).$$

$$= 2yx^4 - 3y^2x^3 + \phi(y)$$

$$\Rightarrow \frac{\partial u}{\partial y} = 2x^4 - 6yx^3 + \phi'(y).$$

We have $\frac{\partial u}{\partial y} = x^2(2x^2 - 6xy) = 2x^4 - 6x^3y.$

$$\therefore 2x^4 - 6yx^3 + \phi'(y) = 2x^4 - 6x^3y$$

$$\Rightarrow \phi'(y) = 0.$$

$$\Rightarrow \phi(y) = c.$$

$$\therefore u = 2yx^4 - 3y^2x^3 + c$$

$$\therefore \text{solution } \boxed{u = 2yx^4 - 3y^2x^3 = c'}$$

Ex. 2 $-y + x \frac{dy}{dx} = 0.$

Sol: Here $M = -y$ and $N = x$

$$\frac{\partial M}{\partial y} = -1 \quad \text{and} \quad \frac{\partial N}{\partial x} = 1$$

$$\Rightarrow \frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}, \quad \therefore \text{the eq is not exact.}$$

$$\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = -1 - 1 = -2$$

$$\frac{1}{M} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = \frac{-2}{-y} = \frac{2}{y} \quad (\text{function of } y)$$

$$\therefore IF = e^{-\int \frac{1}{M} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) dy} = e^{-\int \frac{2}{y} dy}$$

$$= e^{-2 \ln y} = e^{\ln y^{-2}} = \frac{1}{y^2}.$$

Multiply the given eq by $\frac{1}{y^2}$,

$$-\frac{1}{y} + \frac{x}{y^2} \frac{dy}{dx} = 0$$

$$\Rightarrow d\left(-\frac{x}{y}\right) = 0 \Rightarrow -\frac{x}{y} = c$$

$$\Rightarrow \boxed{x = c'y.}$$

Ex. $(e^{x+y} + ye^y) dx + (xe^y - 1) dy = 0, \quad y(0) = -1.$

Ex. $(x^2 + y^2) dx + xy dy = 0.$

Ex $(4xy + 3y^2 - x) dx + x(x + 2y) dy = 0.$

Linear differential equation: (Reducible to exact).

A diff. eq is said to be linear if it is linear in y and $\frac{dy}{dx}$ (i.e., degree one in y and $\frac{dy}{dx}$ and no terms of $y \frac{dy}{dx}$).

Any first order linear diff. eq can be written as

$$\frac{dy}{dx} + P(x)y = Q(x), \quad \text{where } P(x) \text{ and } Q(x) \text{ are function of } x.$$

$$\Rightarrow dy + P(x)y dx = Q(x) dx$$

$$\underbrace{(P(x)y - Q(x))}_{M} dx + \underbrace{1}_{N} dy = 0$$

$$\frac{\partial M}{\partial y} = P(x), \quad \frac{\partial N}{\partial x} = 0$$

The given eq is not exact. (If $P(x) \neq 0$).

$$\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = P(x)$$

$$\frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = P(x) \quad (\text{function of } x)$$

$$\therefore \text{IF} = F = e^{\int \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) dx} = e^{\int P(x) dx}.$$

Multiply the given eq. by $e^{\int P dx}$,

$$e^{\int P(x) dx} \frac{dy}{dx} + e^{\int P(x) dx} P(x)y = e^{\int P(x) dx} Q(x).$$

$$\Rightarrow \frac{d}{dx} (e^{\int P dx} \cdot y) = e^{\int P(x) dx} Q(x)$$

$$\Rightarrow e^{\int P dx} y = \int e^{\int P dx} Q dx + C$$

$$\Rightarrow \boxed{y = \underbrace{e^{-\int P dx} \int e^{\int P dx} Q dx}_{\text{Particular integral}} + C \underbrace{e^{-\int P dx}}_{\text{Complementary function}}}$$

Ex. $y' + y \cot x = 2x + x^2 \cot x.$

Soln This is a linear diff eq.

$$\text{IF} = e^{\int P dx} = e^{\int \cot x dx} = e^{\ln \sin x} = \sin x.$$

$$\therefore \text{solution } y = \frac{1}{\sin x} \int \sin (2x + x^2 \cot x) dx + \frac{C}{\sin x}$$

$$\therefore y = \frac{1}{\sin x} \int 2x \sin x + x^2 \cos x dx + \frac{C}{\sin x}$$

$$\Rightarrow y = \frac{1}{\sin x} \int d(x^2 \sin x) + \frac{C}{\sin x}$$

$$= \frac{1}{\sin x} x^2 \sin x + \frac{C}{\sin x}$$

$$\boxed{y = x^2 + \frac{C}{\sin x}}$$

* Ex. $xy' + 2y = 4x^2, \quad y(1) = 2$

* Ex. $xy' - 2y = x^4.$

Ex. $y' = \operatorname{cosec} x - y \cot x.$

* Ex. $y' - my = e^{mx}.$

Bernoulli eq. (Equation reducible to linear).

$$\frac{dy}{dx} + P(x)y = Q(x)y^n.$$

If $n=0$ or 1 , it is linear.

We can write $\frac{1}{y^n} \frac{dy}{dx} + P \frac{1}{y^{n-1}} = Q(x).$

Let $\frac{1}{y^{n-1}} = z.$

$$\Rightarrow (1-n) \frac{1}{y^n} \frac{dy}{dx} = \frac{dz}{dx}$$

$$\Rightarrow \frac{1}{y^n} \frac{dy}{dx} = \frac{1}{1-n} \frac{dz}{dx}.$$

$$\therefore \left(\frac{1}{1-n}\right) \frac{dz}{dx} + Pz = Q$$

$$\Rightarrow \frac{dz}{dx} + P(1-n)z = Q(1-n)$$

This eq is linear.

Ex. $y' - y = y^2$

Soln $y^{-2}y' - y^{-1} = 1.$

Take $y^{-1} = z \Rightarrow -y^{-2}y' = z'$

$$\Rightarrow y^{-2}y' = -z'.$$

$$\Rightarrow -z' - z = 1$$

$$\Rightarrow z' + z = -1$$

IF = $e^x.$

$$\begin{aligned}
 \underline{\text{Sol}} \quad Z &= e^x \int -1 \cdot e^{-x} dx + c \cdot e^{-x} \\
 &= -e^{-x} e^x + c e^{-x} = -1 + c e^{-x} \\
 \Rightarrow \quad &\boxed{\frac{1}{y} = -1 + c e^{-x}}
 \end{aligned}$$

Ex $(x^2 + 6y^2) dx - 4xy dy = 0.$

Ex $xy' + y = 2x^6 y^4.$

Ex. $6y^2 \frac{dx}{dy} - yx = 2x^4$

Ex $2y dx + x(x^2 \ln y - 1) dy = 0.$

} (Bernoulli in x)